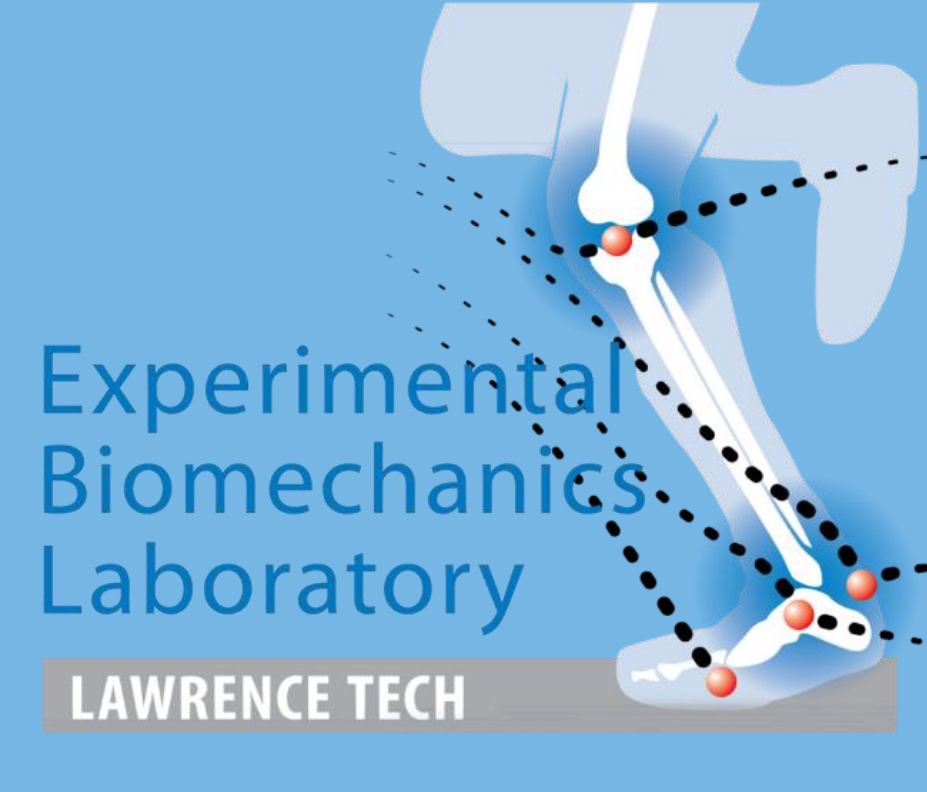
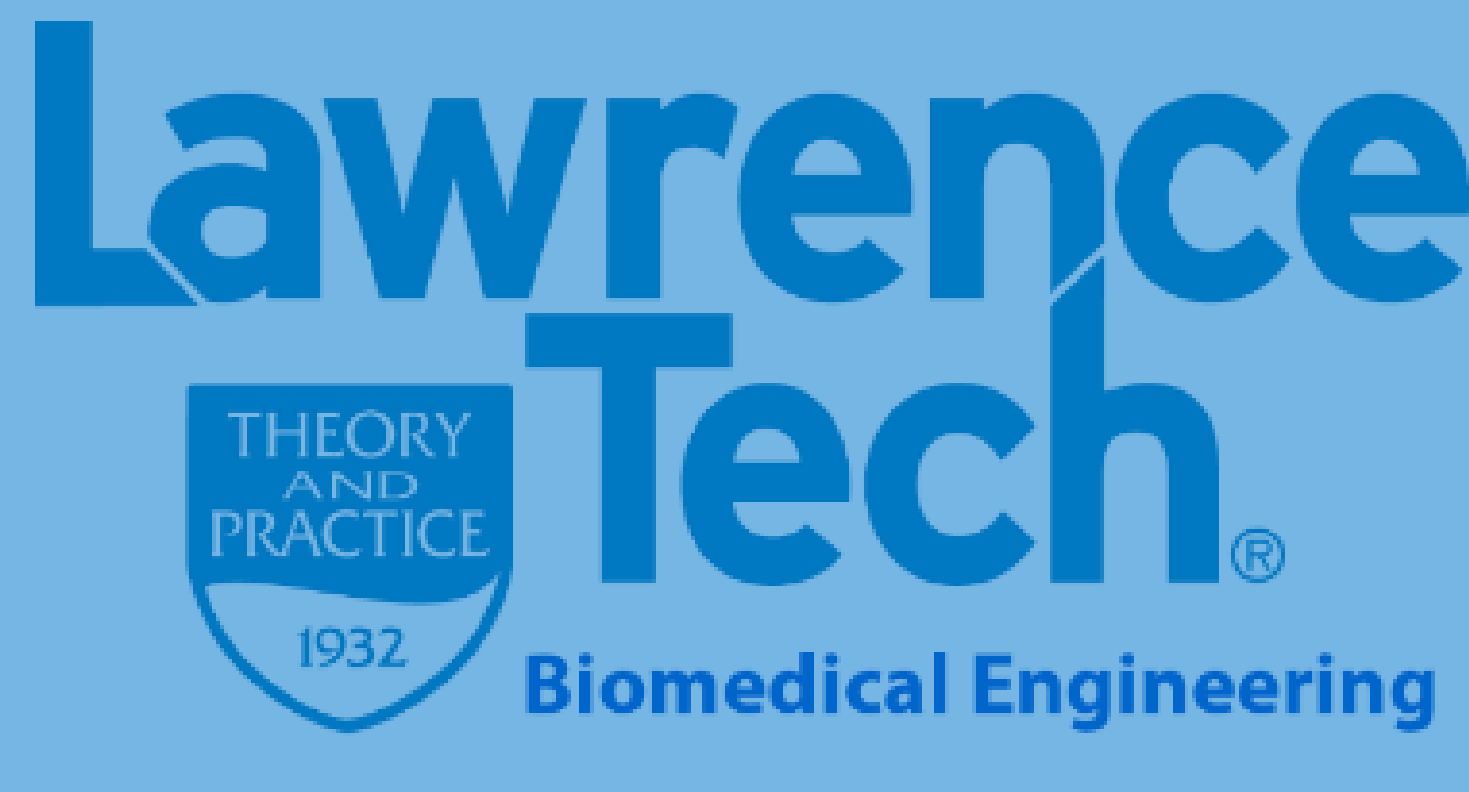


Sustainability Framework Applied To Biomechanics Course Modules To Motivate Opportunities And Collaboration



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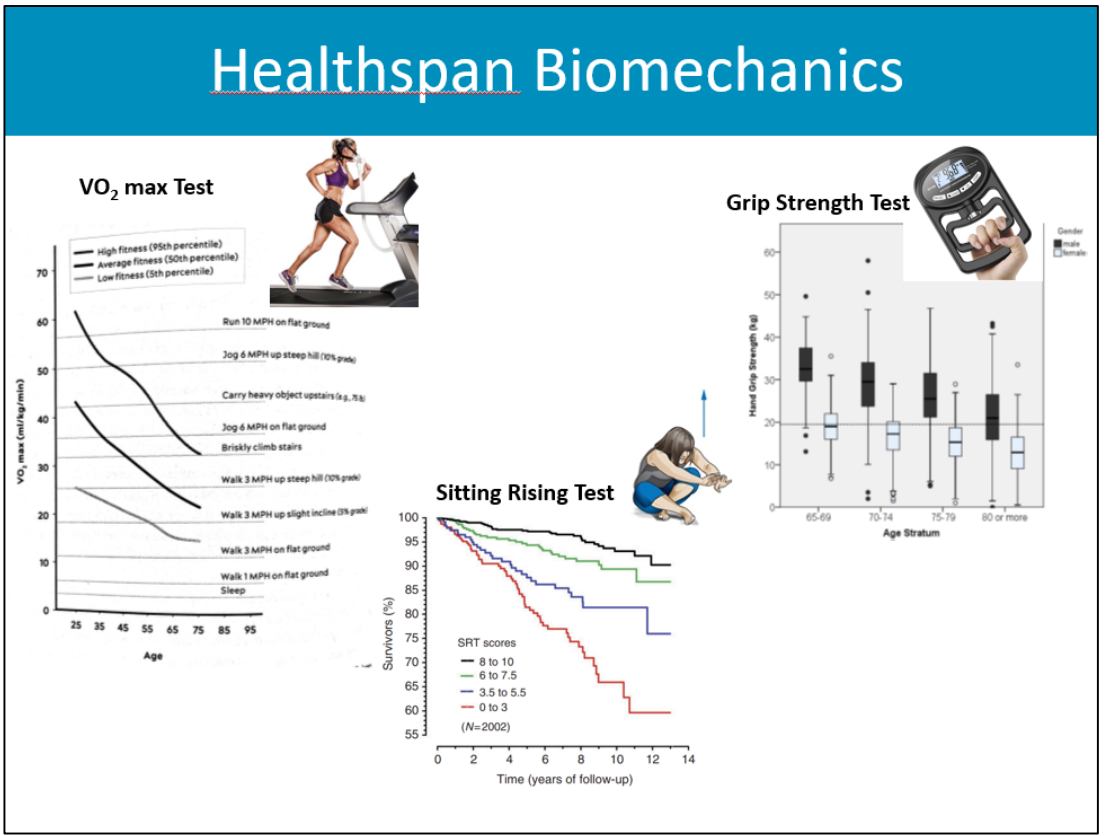
Introduction

- The UN SDGs can be inspiring for students by calling attention to important real-world problems that might lead to opportunities for widescale impact. <https://sdgs.un.org/>

SUSTAINABLE DEVELOPMENT GOALS



- Sustainability frameworks have been applied to biomedical engineering education topics such as; system/network effects, value tensions between stakeholders, empathetic design and strategic foresight.^{1,2}
- SDG #3 “Ensure healthy lives and promote well-being for all at all ages” is relevant to motivate BME 3303 Intro to Biomechanics technical content and multidisciplinary collaborations.³



Project Goals

- To develop course modules that promote opportunities for innovation, social transformation and value creation in biomechanical collaborations between engineers, researchers, designers, clinicians and trainers.

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Methods

- BME 3303 follows Physics and Statics prerequisite courses and covers topics shown below. A design project has been previously shared⁴.

<https://app.engineeringunleashed.com/card/edit/4626>

- Four additional Active, Collaborative Learning modules were developed to relate with specific technical topics while expanding the scope to emphasize non-technical reflections and opportunities to create value for stakeholders.

Semester schedule details

Class	Lecture Topics	In Class Activities	Assignments	
1	Biomechanics Introduction	Video Analysis	Students demo a movement and describe a biomechanical principle	1
2	Statics Review	Quick Think	Joint Anatomy Presentations	
3	Linear Kinematics	Quiz 1	HW	
4	Accelerometry	Quick Think	Use smartphone app phyphox.org to record a daily activity's motion	2
5	Linear Kinetics	Quiz 2	HW	
6	Linear Kinetics Cont.	Grip Strength	Analyze data then compare with Perplexity.ai to create useful insights	3
7	Angular Kinematics	Quiz 3	HW	
8	Group Design Project	Sport Analysis	Ideation, design selection, testing and value for athletes/coaches	
9	Inertial Measurement	Quiz 4	Setup of mBient 3D gyrometer sensor and calibration	
10	Work-Energy	Quick Think	HW	
11	Impulse-Momentum	Quiz 5	HW	
12	Exam 1 Review	Project Help		
13		E1 Dynamics	Midterm Exam 1	
14	Uniaxial Deformation	Project Presentations	HW	
15	Stress and Strain	Quick Think	Tissue Properties Presentations	
16	Material Properties	Tension Testing	HW	
17	Poisson's Ratio	Quick Think		
18	Stress Transformation	Demonstration	HW	
19	Mohr's Circle	Quick Think		
20	Failure	Examples	HW	
21	Torsion	Quick Think		
22	Bending	Examples	HW	
23	Bending Stress	Quick Think		
24	Combined Loading	Examples		
25	Exam 2 Review			
26		E2 Deformation	Midterm Exam 2	
27	Stability and Walking	Sit-to-Stand	Use sit2stand.ai then expand the assessment to stakeholder groups	4
28	Running	Examples	HW	
29	Engineered Running	Debate		
30	Final Exam Review			
Final			Cumulative Final Exam	

1 Biomechanical video analysis

<https://app.engineeringunleashed.com/card/edit/4619>

2 Center of gravity tracking with an IMU

<https://app.engineeringunleashed.com/card/edit/4618>

3 Grip strength population variations

<https://app.engineeringunleashed.com/card/edit/4605>

4 Sit-to-stand functional analysis⁵

<https://app.engineeringunleashed.com/card/edit/4297>

Results

- Hands-on activities using video analysis, IMUs and force sensors were brought into the classroom to support lecture topics instead of as laboratory activities.

Reflection prompt examples for modules

1 Identifying Biomechanical Concepts

- Biomechanics Analysis
 - ESPN Sports Science Episodes
 - Fastest Lacrosse Shot
 - Ndamukong Suh

What are the important parameters that are discussed?

Conduct your own biomechanical analysis of videos... (Homework #1)

Eds Meyer (CERN)

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Hallway walking trial of approximately 29 steps in 17 seconds (1.7 steps/sec).

3 axis accelerometer data (with gravity). X-direction: +Right/-Left, Y-direction: +Up/-Down, Z-direction: +Forward/-Backward.

The SDs for each direction during the 17 sec walking were X = 1.02, Y = 2.79 and Z = 1.12 (m/s²). This makes sense that there were larger accelerations in the vertical direction and the smallest in the side-to-side directions. The distance covered during the trial should have been measured or GPS tracking could have been collected at the same time to also get velocity.

The vertical accelerometer could be used to measure ground reaction forces by knowing the mass of the person and using F=ma.

3 Quantitative Analysis

- Use the Hand Dynamometer to measure grip strength on both hands (3 trials for each).

Handexer Dynamometer

<https://www.youtube.com/watch?v=1B8N6aTupc>

4 Creating Value

- How did you do?
 - Download your detailed results.
 - What are some ways that you could improve?
- What are some other uses for this technique?
 - Think about other groups...

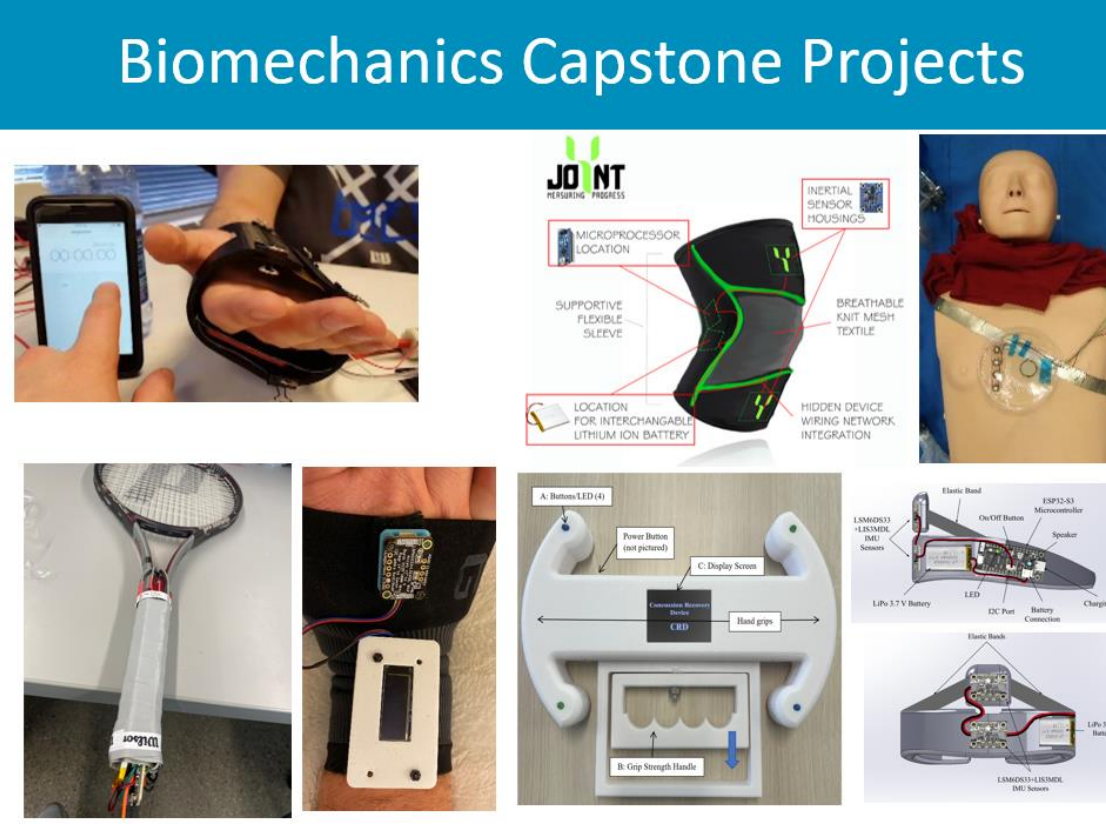
<https://askandai.com/assets/results/Cash2025/>

- Students were able to use the measurement and analysis techniques. Feedback showed that they enjoyed the hands-on aspects of the modules and the open-ended, real world nature of the projects.
- Students gained experience and perspective for human biomechanics, engineering design, innovation and sustainability goals.

Discussion

- The real-world theme supported student learning of biomechanical concepts.
- Supports LTU “Theory & Practice” and the Entrepreneurial Mindset.
- Modules limited in duration didn’t “solve” the problems of sustainability for human health and well-being.

- The ultimate goal is to support students’ strategies for making informed decisions in engineering or other disciplines.



References: [1] Warren et al. (2010), J Sustainability Ed 6; [2] Tranquillo (2018), ASEE Proc; [3] Hu et al. (2023), Sustainability 15(11); [4] Meyer and Ulrey (2016), ASEE Proc; [5] Boswell et al. (2023) Digit Med 6(32).

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